

Bill Of Materials for ACQ-02_M90E36A_3P4W_2025_0422

Design Title ACQ-02_M90E36A_3P4W_2025_0422
Author Gustavo E. Real
Document Number
Revision
Design Created martes, 22 de abril de 2025
Design Last Modified martes, 22 de abril de 2025
Total Parts In Design 172

0 Modules

Quantity	References	Value	Description
Sub-totals:			
38 Capacitors			
Quantity	References	Value	Description
12	C1, C2, C3, C4, C6, C31, C32, C34, C35, C36, C37, C38	0.1u	0.1µF ±10% 100V Ceramic Capacitor X7R 1206 (3216 Metric) SMD
2	C5, C7	10u	10µF ±10% 6.3V Ceramic Capacitor X7R 1206 (3216 Metric) SMD
22	C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29	0.018u	0.018µF ±10% 50V Ceramic Capacitor X7R 1206 (3216 Metric) SMD
1	C30	470u	CAP TANT 470uF 10V 10% LOESR SMD
1	C33	100u	100µF 6.3V 20% Aluminum Polymer Capacitor 2917 (7343 Metric) SMD 18mOhm

Sub-totals:

66 Resistors

Quantity	References	Value	Description
1	R1	10.0k	10 kOhms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) SMD Thin Film
5	R2, R65, R66, R67, R68	330	330 Ohms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) SMD Thin Film
8	R6, R7, R8, R9, R10, R11, R12, R13	510	510 Ohms ±5% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) SMD Thick Film
22	R14, R15, R16, R24, R32, R40, R41, R42, R45, R46, R49, R50, R53, R54, R57, R58, R59, R60, R61, R62, R63, R64	1.00k	1 kOhms ±1% 1/4W Chip Resistor 1206 (3216 Metric) SMD Thin Film
21	R17, R18, R19, R20, R21, R22, R23, R25, R26, R27, R28, R29, R30, R31, R33, R34, R35, R36, R37, R38, R39	240k	240 kOhms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) SMD Thin Film
8	R43, R44, R47, R48, R51, R52, R55, R56	2.43	2.43 Ohms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) SMD Thick Film
1	R69	510	RES 510 OHM 1/8W 5% 0805 (2012 metric) SMD

Sub-totals:

9 Integrated Circuits

Quantity	References	Value	Description
1	U1	TCMT4100	Optoisolator Transistor Output 3750Vrms 4 Channel 16-SOP
1	U2	TMLM04-105	Enclosed AC DC Converter 1 Output 5V 800mA 90 ~ 264 VAC, 120 ~ 370 VDC Input
1	U3	LM1117MP-3,3	Linear Voltage Regulator NCP1117 IC Positive Fixed 1 Output 1A SOT-223
1	U4	25LC512-I_SN	512k,64K X8 , 2.5V Serial EE SOIC-8
5	U5, U6, U7, U8, U9	FODM8071	Logic Output Optoisolator for SPI 20Mbps Push-Pull, 3750Vrms 1 Channel 20kV/µs CMTI 5-Mini-Flat - 6-SOIC

Sub-totals:

0 Transistors

Quantity	References	Value	Description
Sub-totals:			

5 Diodes

Quantity	References	Value	Description
4	D1,D2,D3,D4	LED Rojo	LED Rojo 645nm - Discrete 1.9V 0603 (1608 Metric)
1	D5	LED Verde	LED Verde 645nm - Discrete 1.9V 0603 (1608 Metric)

Sub-totals:

54 Miscellaneous

Quantity	References	Value	Description
15	3.3V,I1N,I1P,I2N,I2P,I3N,I3P,I4N,I4P,J7,J8,V1P,V2P,V3P,VN	CONN-SIL1	1 Position Male Header, Breakaway Connector, Contact length 0.318, Through Hole, Gold
1	IC1	M90E36A	ATMEL 3 Phase Meter IC ATM90E36A 48-TQFP (7x7) - 1.8V/3V
1	J1	CONN-DIL8	8 Positions Male Header, Two files, Breakaway Connector, 0.100 (2.54mm), Contact length 0.318, Through Hole, Gold
1	J2	CONN-DIL14	14 Positions Male Header, Two files, Breakaway Connector, 0.100 (2.54mm), Contact length 0.318, Through Hole, Gold
1	J3	CONN-SIL8	8 Positions Male Header, One file, Breakaway Connector, 0.100 (2.54mm), Contact length 0.318, Through Hole, Gold
2	J4,J5	TBLOCK-I8	8 Position Wire to Board Terminal Block Horizontal con placa 0.200 (5.08mm) Orificio pasante
1	J6	TBLOCK-M2	2 Position Wire to Board Terminal Block Horizontal con placa 0.197 (5.00mm) Orificio pasante
3	JP1,JP2,JP3	JUMPER	2 Positions Male Header, One file, Breakaway Connector, 0.100 (2.54mm), Contact length 0.318, Through Hole, Gold
1	JP4	JUMPER	Animated Jumper
24	JP5,JP6,JP7,JP8,JP9,JP10,JP11,JP12,JP13,JP14,JP15,JP16,JP17,JP18,JP19,JP20,JP21,JP22,JP23,JP24,JP25,JP26,JP27,JP28		Thin Film Resistors - SMD MCT 0603 Jumper VG01 P5
3	VR1,VR2,VR3	S05K250	Varistor, Leaded standard series, V=250Vrms(320Vdc), Imax=400A(8/20us), W=8.2J, Pmax=0.1W
1	X1	CRYSTAL	16.384MHz ±20 ppm Cristales 10pF HC-49/US

Sub-totals:

Totals:

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